Eliminating Excess Gap of Spoiler to Roof on Scorpio Vehicle.

Mahindra and Mahindra Ltd, Scorpio Body Shop, Nasik Plant

	Plan	Actual
Start Date	**03th Jan-12**	**04th Jan-12**
Completion Date	**31st March-12**	**31st May-12**

Current Status:

Closed

Cross Functional Team

Team Leader – Leads the team, Co-ordinates the meeting
Mentor - Facilitates the team for concern resolution, provides resource.
Mostly Mentor is PU Manager.
Members – Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	**N.S.Hiray**	Deputy Manager	MQA,(SBS)	9881737077	Hiray.nitin2@mahindra.com
2	**Mentor**	**R.G.Suryavanshi**	Dept.Head	SBS	9881378988	SURYAVANSHI.RAVINDRA@mahindra.com
3	**Member 1**	**Mangesh Gangurde**	Asst. Manager	MQA	9922644399	GANGURDE.MANGESH@mahindra.com
4	**Member 2**	**Vivek Bhat**	Deputy Manager	Tooling	9922438403	BHAT.VIVEK@mahindra.com
5	**Member 3**	**Manoj More**	Deputy Manager	Production	9922433518	MORE.MANOJ@mahindra.com

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No	Description	Name	Designation	Dept	Phone	Email
6	**Member 4**	**Chaitanya Dasakkar**	Asst. Manager	MAPS	9420901559	DASAKKAR.CHAITANYA2@mahindra.com
7	**Member 5**	**S.Y.Deshpande**	MANAGER,QA	MQA	9422770889	DESHPANDE.SANDEEP2@mahindra.com
8	**Member 6**	**A.S Hasme**	MANAGER,TOOLING	Tooling	9822248250	HASME.ASHUTOSH@mahindra.com

1. Select The theme

Problem Definition :

What: **Excess gap Spoiler to Roof**

Where: **Concern Reported in daily CAI and TCF report**

Who: **Concern reported by internal customer TCF and as well as in CAI and WRAP**

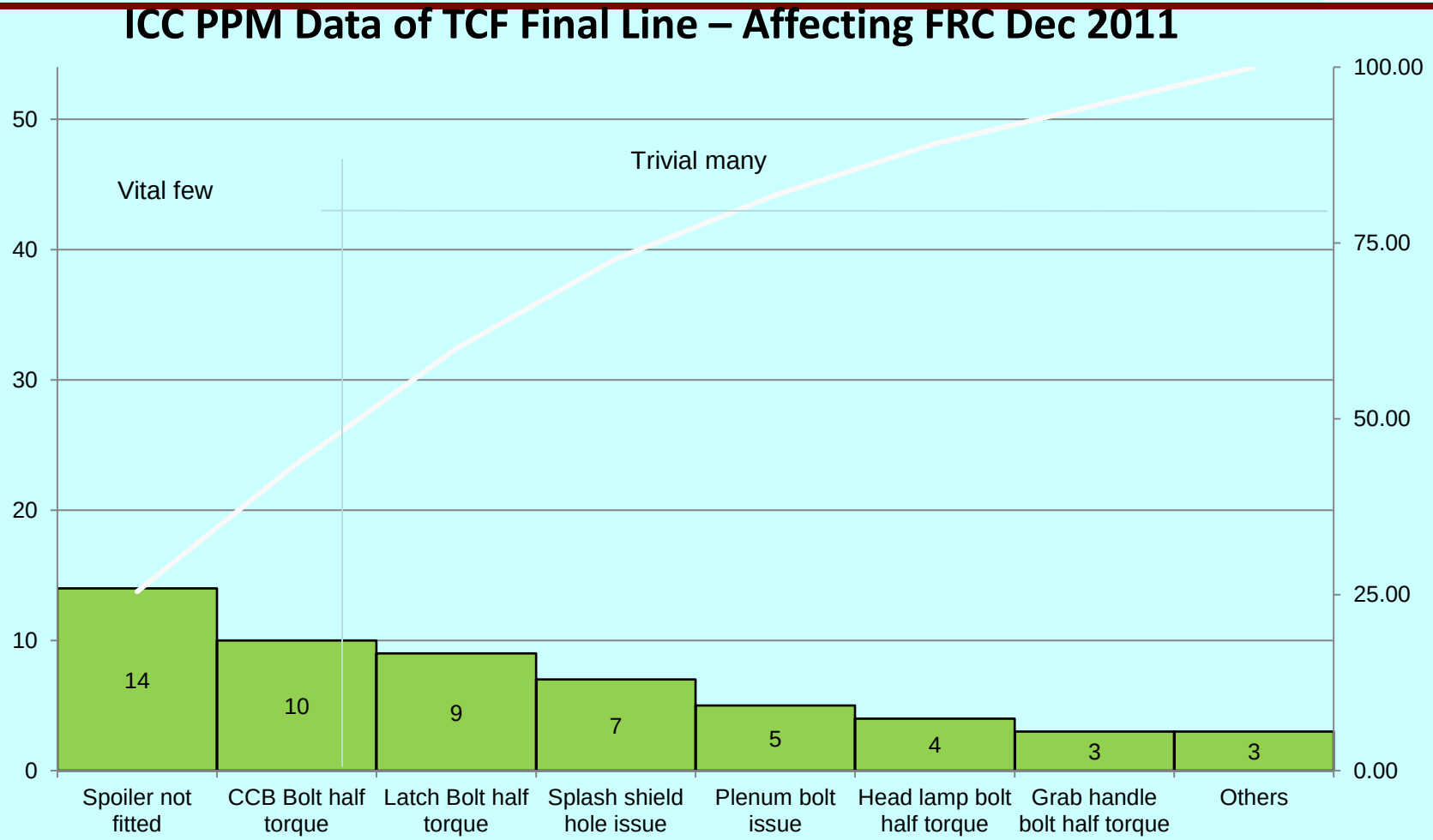
When: **since December '11**

Why: **To be investigated**

How: **It is seen visually clearly**

Spoilar bolt mismatch with rear header hole

2. Justification



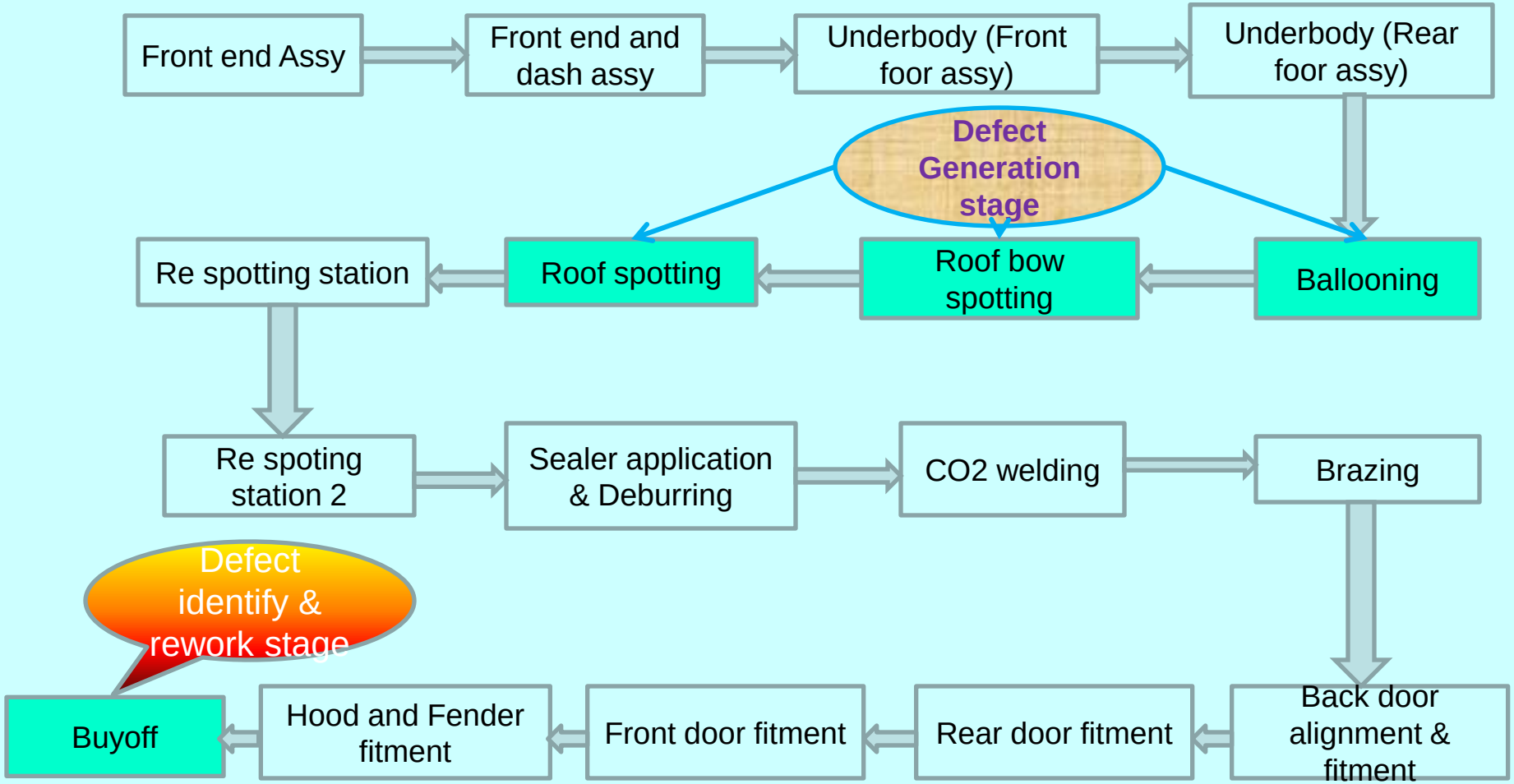
Cost of Poor Quality = **9590 Rs** for the month of Dec-11

2.a) Immediate Containment Action

Step Pin is Made with Roof to Rear Header Hole Concentricity so that it will ensure correct location of spoiler at TCF

3. Understand the Current Situation

Process Flow – TOP VIEW



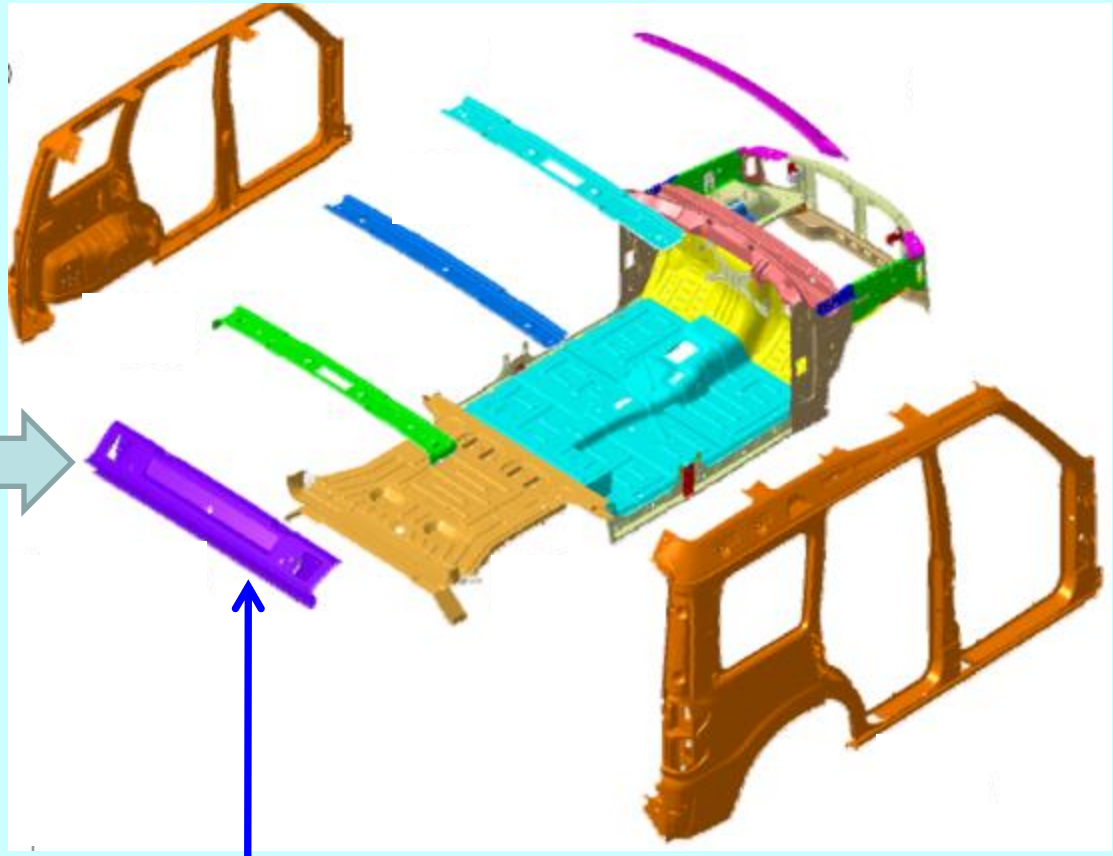
3. Understand the Current Situation

Fitment Details of Defect Stage (Ballooning)

RH side Assy

Ballooning

LH side Assy



Rear Header assy
By M/S MUSCO

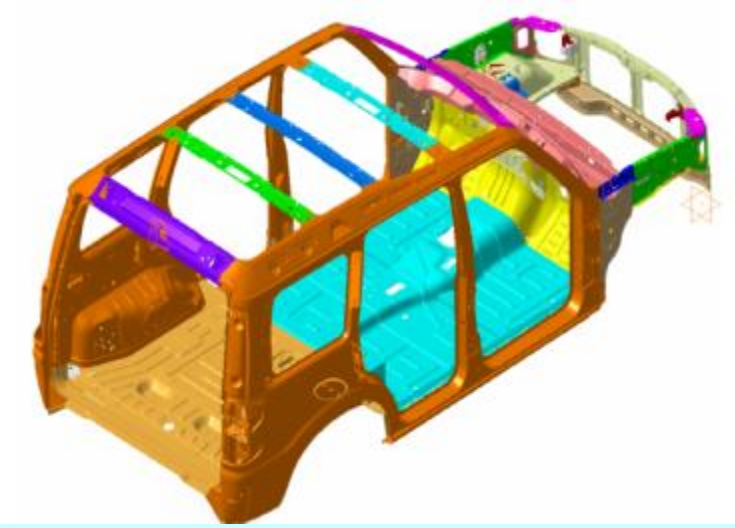
3. Understand the Current Situation

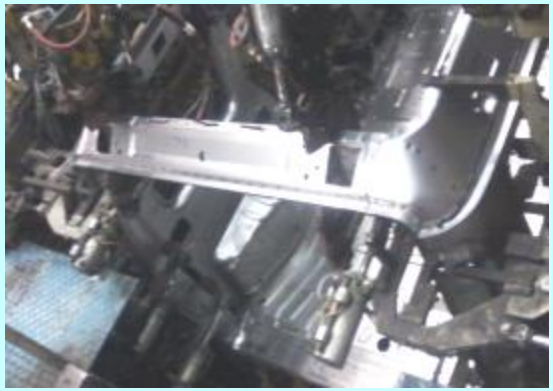
Fitment Details of Defect Stage (Framing)

RH side Assy

Ballooning → Framing

LH side Assy





At Framing station rear Header positioning done by locating pins and clamps

3. Understand the Current Situation

Fitment details of Defect Stage (Roof Assy)

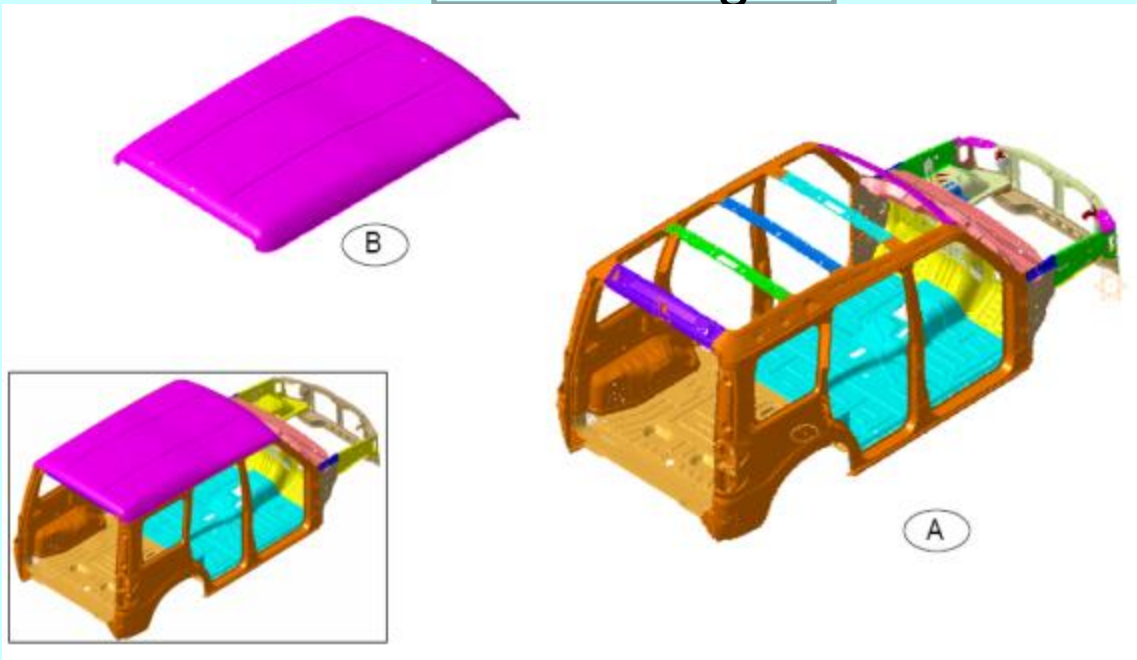
RH side Assy

Ballooning

LH side Assy

Framing

Roof Framing

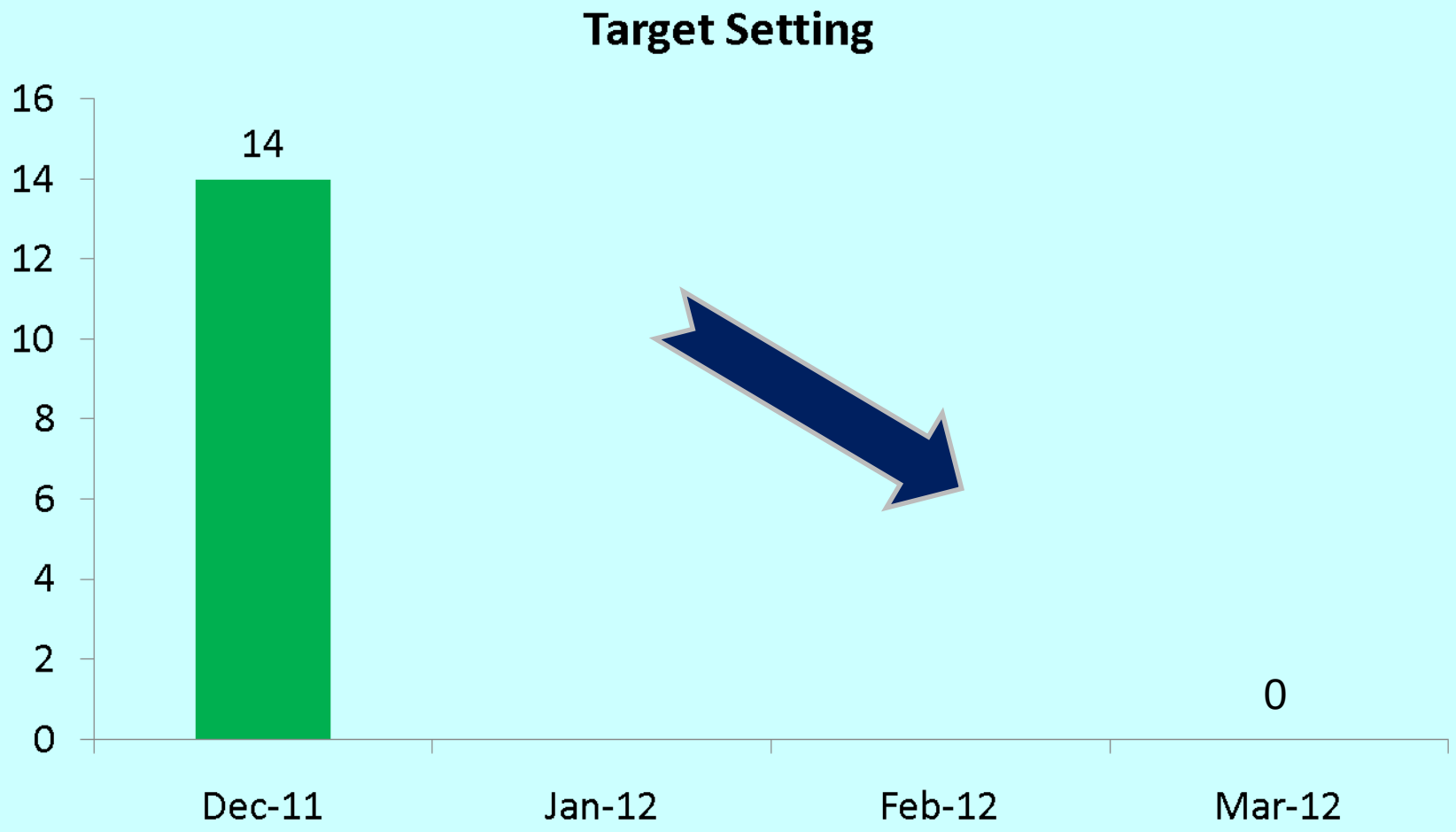


3. Understand the Current Situation

CFT Observations on GEMBA

- Rear header Assy Mismatch was found on 100% assemblies.
- At Framing stn#20, Rear header positioning pin was not inserting into some bodies.
- At Framing stn# 40 Roof location clamp profile not matching with roof causing incorrect roof location.
- Due to spot on rear header extension bracket rear header will not sit on D pillar assy.
- At Framing stn#20 rear header positioning not possible due to drip rails overlap.

4. Select Target



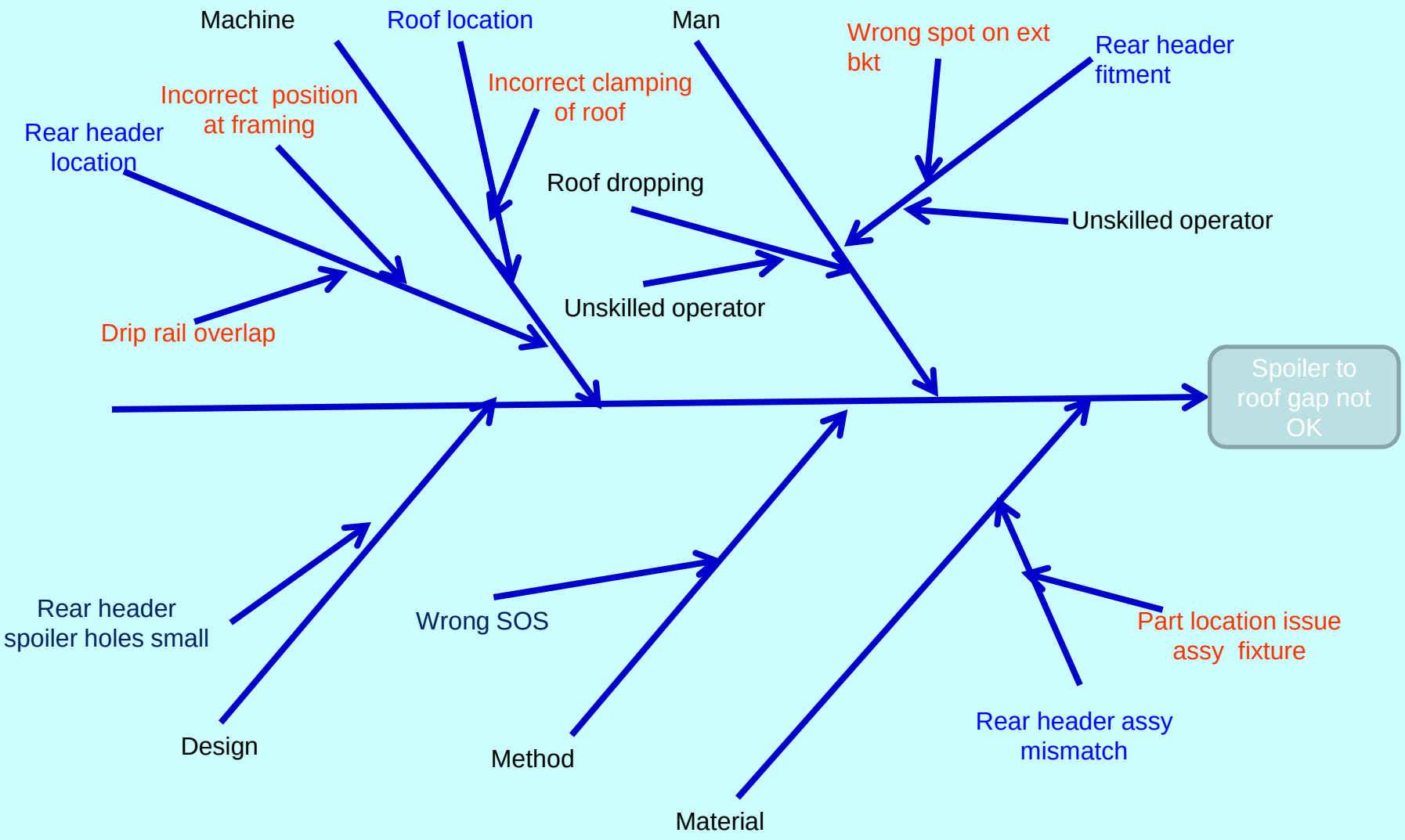
5. Analysis

5.1 : Identify all Possible Causes

- Rear header positioning not ok due to Drip rails Mismatch/Overlap at framing stn#20 .
- Rear header will not sit on D pillar assy due to spot on rear header extension bracket.
- Incorrect dropping of roof at Roof assy Stage.
- SOS not followed by stage operators.
- Incorrect Roof location due to clamp profile not matching with roof at Framing.
- Rear header Assy Mismatch will causing excess gap.
- Rear header positioning pin was not inserting properly during spot welding operations at framing stage.

5. Analysis

5.2 : Identify Possible causes: Cause and Effect Diagram



5. Analysis

5.3 : TEST OF HYPOTHESIS: Confirm the probable Cause.

Sr No	Probable cause	Testing and Observation	Conclusion
1	Rear header assy Mismatch	We checked part at M/S MUSCO on checking gauge in which it was found hole mismatch	Valid

Total 20 Assemblies are check at vendor End

Sr.No	Rear Header Assy	Mismatch observed
1	Assy-01	Yes
2	Assy-02	Yes
3	Assy-03	Yes
4	Assy-04	Yes
5	Assy-05	Yes
6	Assy-06	Yes
7	Assy-07	Yes
8	Assy-08	Yes
9	Assy-09	Yes
10	Assy-10	Yes

Sr.No	Rear Header Assy	Mismatch observed
11	Assy-11	Yes
12	Assy-12	Yes
13	Assy-13	Yes
14	Assy-14	Yes
15	Assy-15	Yes
16	Assy-16	Yes
17	Assy-17	Yes
18	Assy-18	Yes
19	Assy-19	Yes
20	Assy-20	Yes

5.3 : TEST OF HYPOTHESIS: Confirm the probable Cause.

Sr No	Probable cause	Testing and Observation	Conclusion
2	Rear header location at Framing incorrect due to locating pin	At Framing stn#20 rear header positioning pin was not inserting into some bodies	Valid

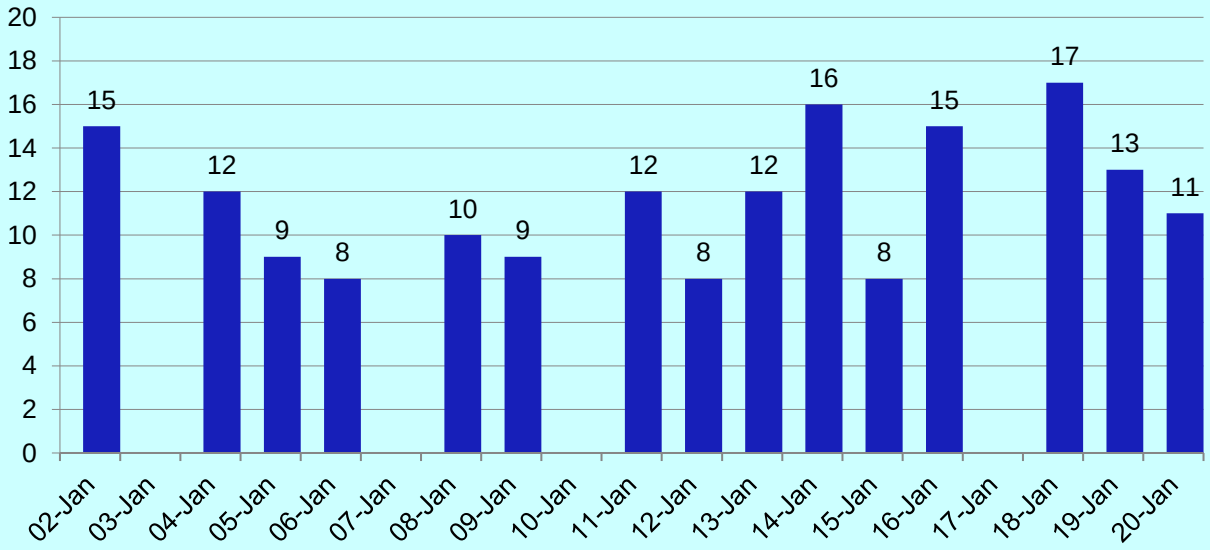
Sr.No	Date	BIW Number	Loc pin of Rear Header Inserted properly
1	2-Dec	50532	Yes
2	2-Dec	50533	Yes
3	2-Dec	50534	Yes
4	2-Dec	50535	No
5	2-Dec	50536	Yes
6	2-Dec	50537	Yes
7	2-Dec	50538	No
8	2-Dec	50539	Yes
9	2-Dec	50540	Yes
10	2-Dec	50541	NO
11	2-Dec	50542	Yes
12	2-Dec	50543	Yes
13	2-Dec	50544	Yes
14	2-Dec	50545	Yes
15	2-Dec	50546	Yes

Sr.No	Date	BIW Number	Loc pin of Rear Header Inserted properly
16	4-Dec	50712	No
17	4-Dec	50713	Yes
18	4-Dec	50714	Yes
19	4-Dec	50715	Yes
20	4-Dec	50716	No
21	4-Dec	50717	Yes
22	4-Dec	50718	NO
23	4-Dec	50719	Yes
24	4-Dec	50720	Yes
25	4-Dec	50721	Yes

5.3 : TEST OF HYPOTHESIS: Confirm the probable Cause.

Sr No	Probable cause	Testing and Observation	Conclusion
3	Rear header location at Framing incorrect due to drip rail overlap	At Framing stn#20 rear header positioning not possible due to drip rails overlap	Valid

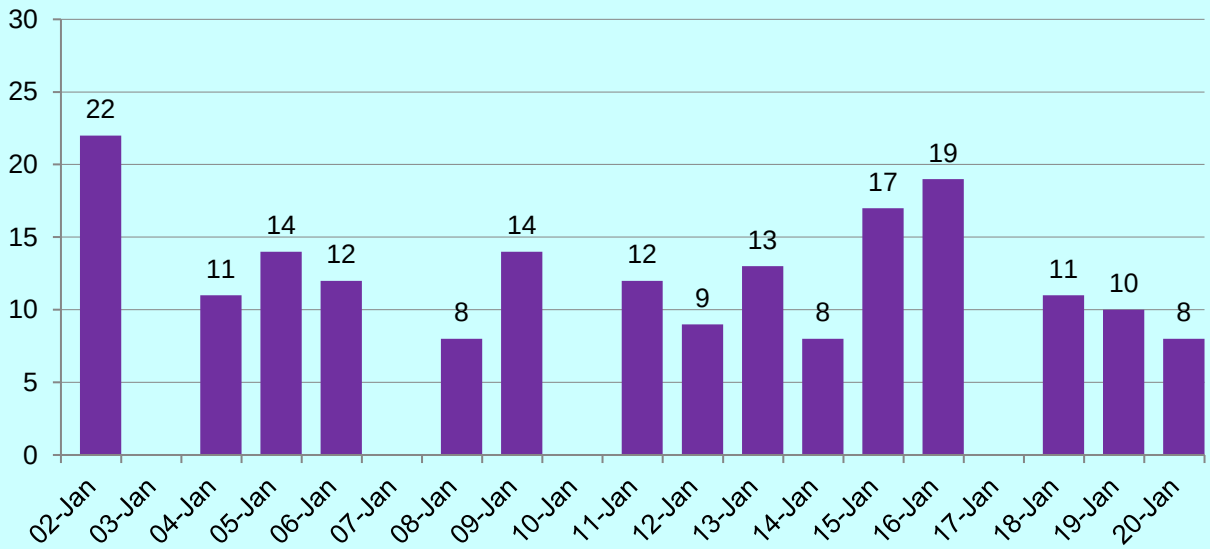
Day wise Defect Monitoring of Drip Rail Overlap.



5.3 : TEST OF HYPOTHESIS: Confirm the probable Cause.

Sr No	Probable cause	Testing and Observation	Conclusion
4	Roof location incorrect	At framing stn# 40 roof location clamp profile not matching with roof causing incorrect roof location	valid

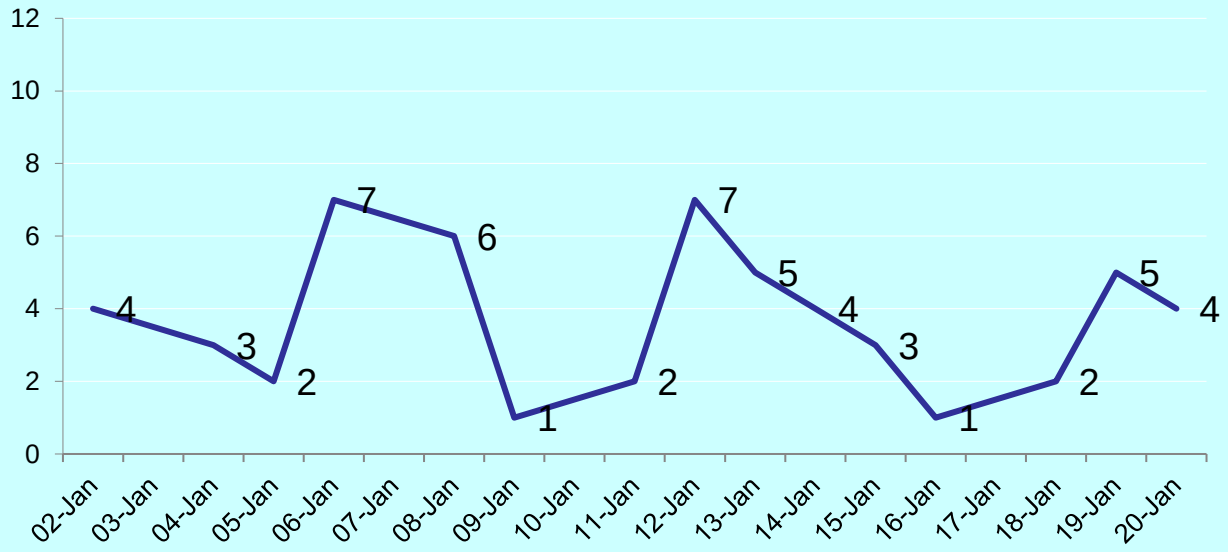
Day wise Monitoring of Roof location incorrect.



5.3 : TEST OF HYPOTHESIS: Confirm the probable Cause.

Sr No	Probable cause	Testing and Observation	Conclusion
5	Rear header fitment at Ballooning not possible	Due to spot on rear header extn bkt rear header will not sit on D pillar	Valid

Day wise Monitoring of Rear header fitment .



5. Analysis

5.4 : Why Why Analysis of Possible Causes

Probable cause	why	why	why	why	why
Rear header assy mismatch	Incorrect location on assy fixture	Part location pin found loose	Bolting of pin was loose		
Rear header location at Framing incorrect	Rear header locating pin was not inserting	Rear header assy was found mismatch	Incorrect location on assy fixture	Part location pin found loose	Bolting of pin was loose
Rear header location at Framing incorrect	Rear header locating pin was not inserting	RH drip rails found overlap			
Roof location incorrect	Roof location clamping incorrect	Roof location clamp profile not matching with roof	Clamp profile worn out		
Rear header fitment at Ballooning not possible	Spot taken on rear header ext bkt and D plr opening area	Operator error			

6. Implement Corrective Measures

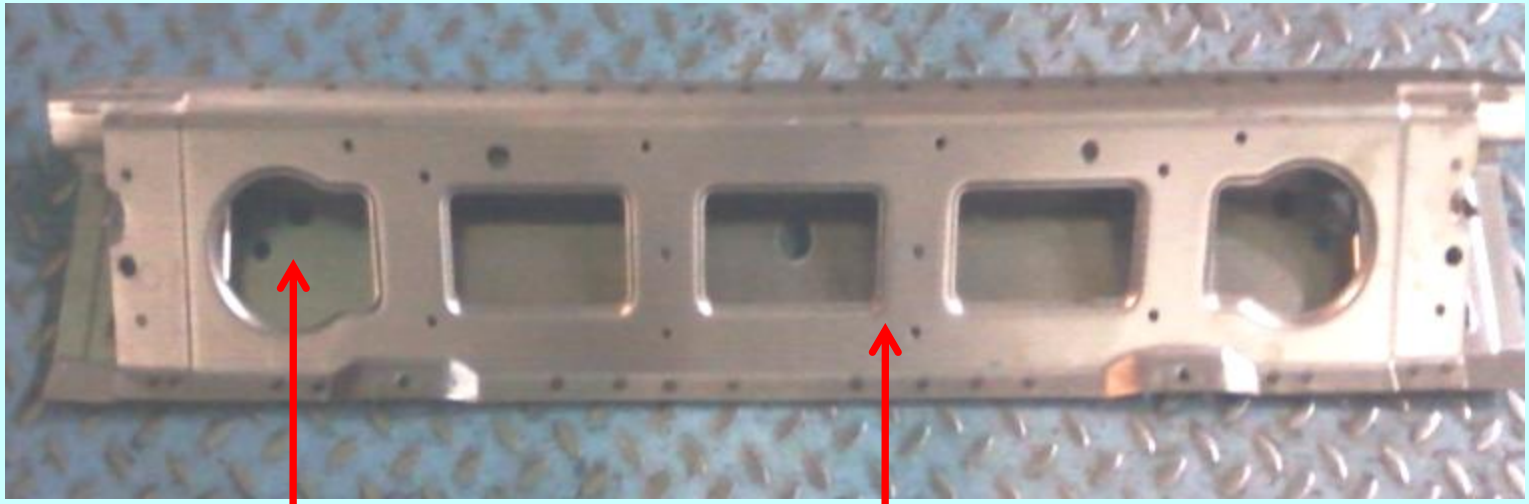
Root cause	Action plan	Target date	Resp.	Status
Rear header assy found mismatch	Part location pin found loose which needs to be tighten	05.01.2012	Vivek/Gokul	Closed
Rear header location at Framing incorrect	Rear header mismatch to be restored	05.01.2012	Vivek/Gokul	Closed
Rear header location at Framing incorrect Due to drip rails overlap	RH side drip rail to be cut by 2 mm	27.01.2012	Giri D	Closed
Roof location incorrect due to roof clamp profile worn out	Roof location clamp profile matched with roof and additional stopper provided to set the roof	01.02.2012	Vivek	Closed
Rear header fitment at Balloning not possible	RH side operators are given DRF that not to take spot on ext bkt and D plr top opening area	25.01.2012	Deshmukh K	Closed

6. Implement Corrective Measures

Root cause	Action plan	Target date	Resp.	Status
No checkpoint for spoiler hole location	Checkpoint is added in IQ check sheet	22.01.2012	Mangesh	Closed
Hole of rear header is smaller than roof hole	Hole size of rear header to be increased from 12 mm to 16 mm	10.02.2012	Sunil Deshmukh	Closed.

6. Implement Corrective Measures

1.Rear header assy mismatch reduced by tightening the part location pin at M/S MUSCO

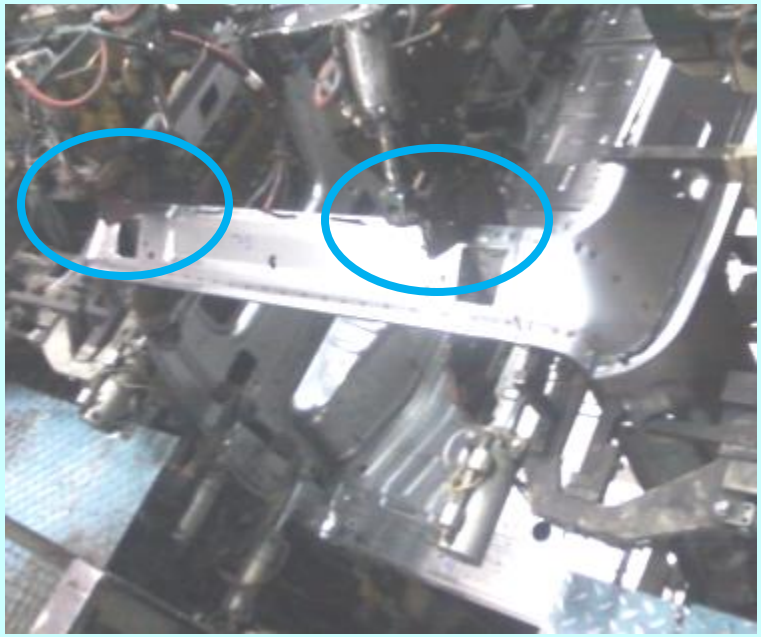


Outer pnl having spoiler hole was shifted wrt inner panel

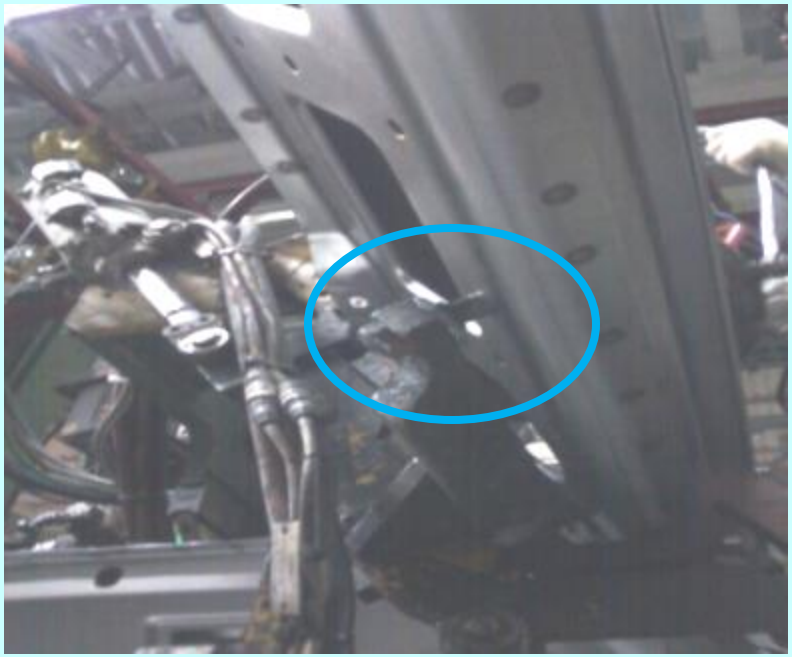
Inner panel

6. Implement Corrective Measures

2.Rear header assy located correctly at framing station by eliminating assy mismatch at M/S MUSCO



Rear header clamping at Framing Stn # 20



Rear header locating correctly at Framing Stn # 20

6. Implement Corrective Measures

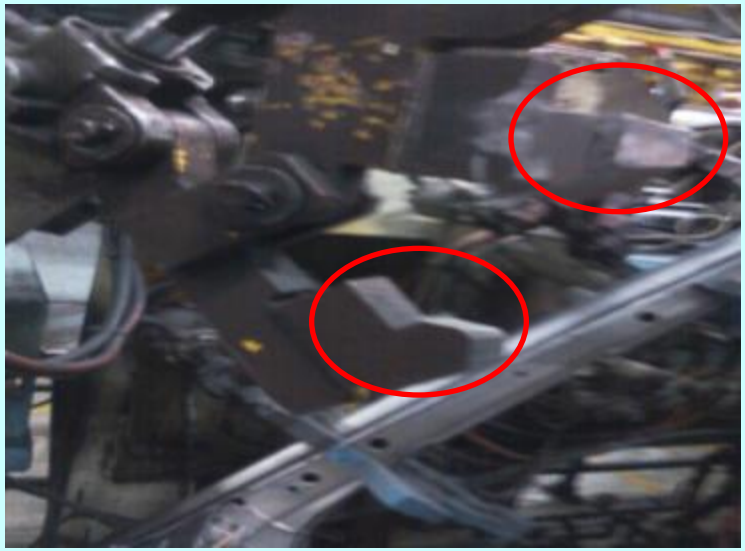
3.Rear long drip rail cut by 2mm to avoid drip rail overlap by M/S Star Auto

Drip rail cut
by 2mm

6. Implement Corrective Measures

4. Roof location clamp profile matched with roof and additional stopper provided to set the roof to avoid incorrect roof location

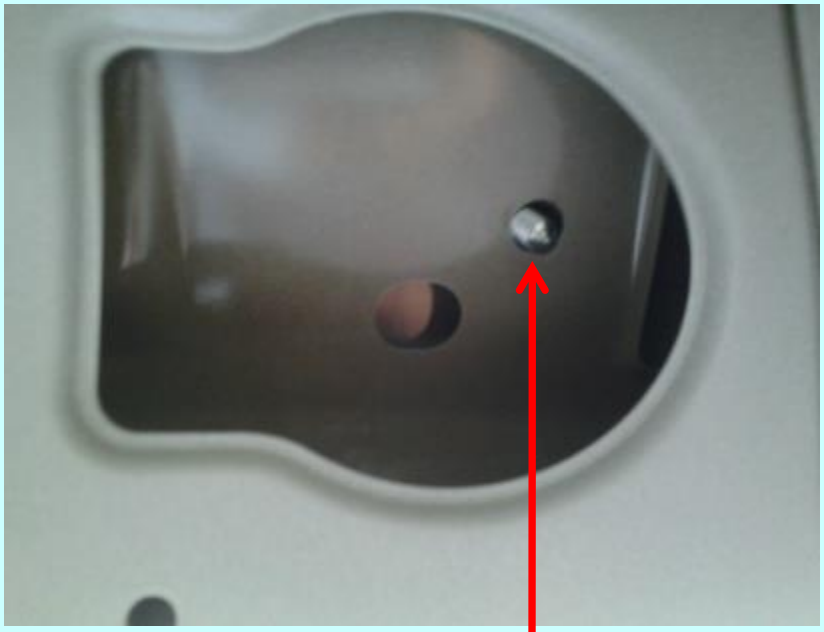
Additional stopper for roof location



Roof clamp profile matched with roof profile

6. Implement Corrective Measures

5. Spoiler mounting hole size to be increased from 12mm to 16 mm to compensate the total variation



Spoiler mounting hole on rear header outer panel

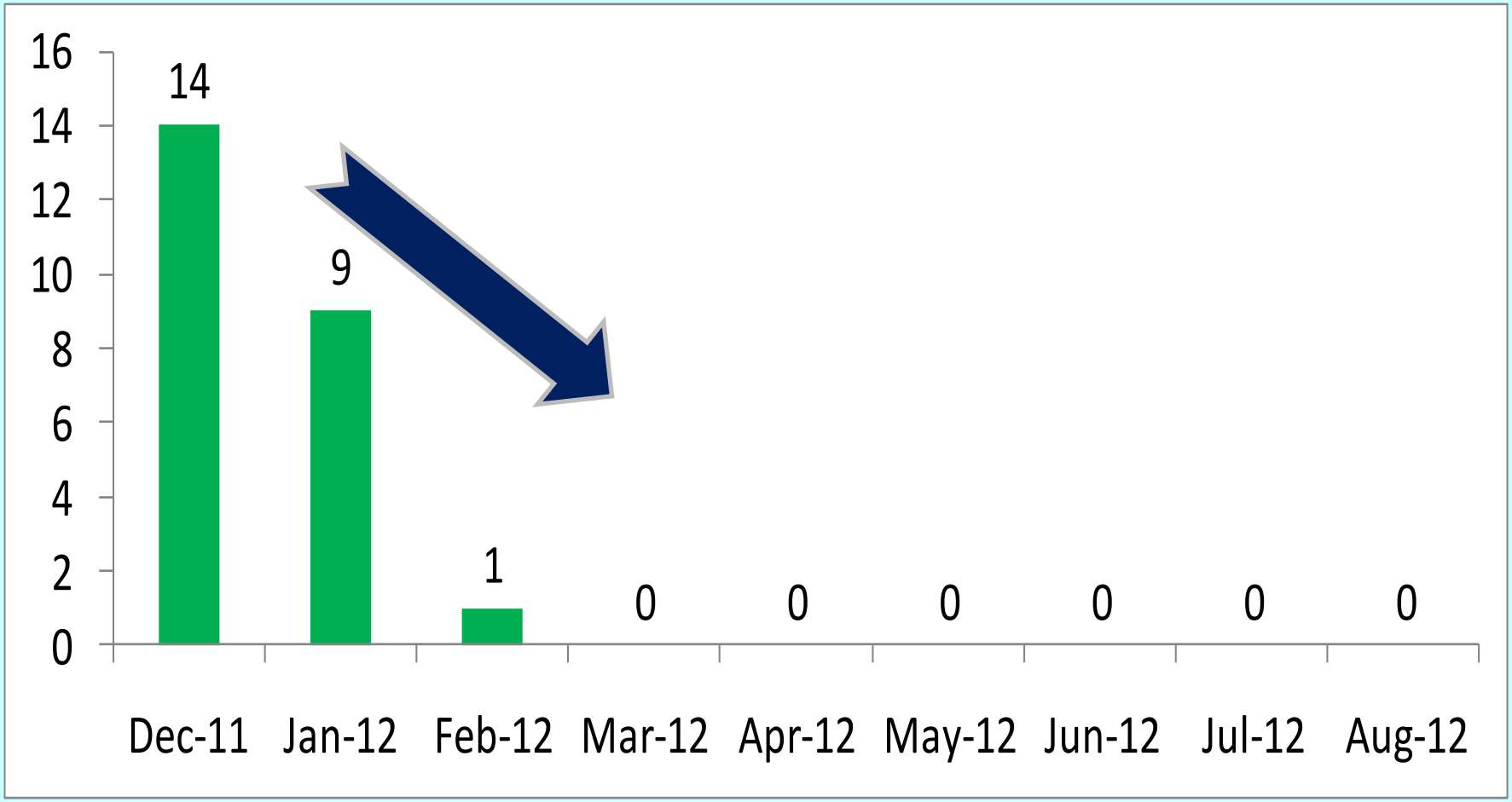
6. Implement Corrective Measures

5. Checkpoint is added in IQ check sheet for daily monitoring of spoiler mounting hole position

IQ-CHECK LIST -SCORPIO BODY SHOP

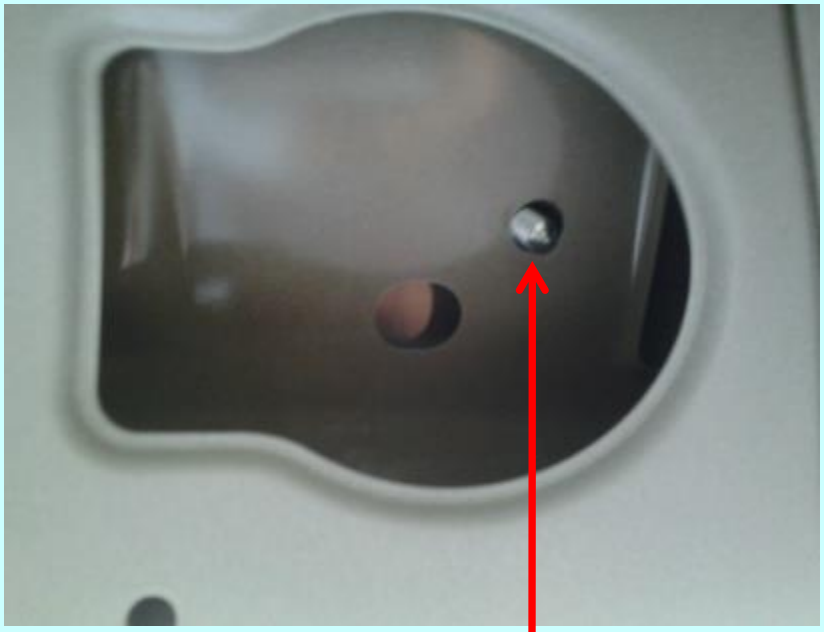
Sr No	BWT No	BWT Name	Phenomena	Physical Phenomena
1	BWT 1	Underbody (Plenum,Dash,Front end to Rear floor) Z-02,Z-14,Z-03,Z-05,Z-06	No of studs	Dash panel
			Mismatch	Hole mismatch accelerator mtg. bkt
			Damage	Head lamp mounting bkt.
			Mismatch	Front seat mounting channel hole
			Spot twist	Hand bracket area
			Burr	Scuff plate holes
2	BWT 2	Framing line(Balloning to Deburring) Z-08+Z22	Mismatch	Drip rails front and rear
			Locating	Rear header on framing station
			Overlap / Excess gap	Roof on A pillar
			Mismatch	Spoilar and Ski rack
			Burr removal	LH/RH A pillar,Spot on edge -D pillar closure
			Damage	Roof LH/RH

7.1 Check Result



7.2 Full Scale Implementation Plan

Spoiler mounting hole size to be increased from 12mm to 16 mm to compensate the total variation Target date – 30.03.2012

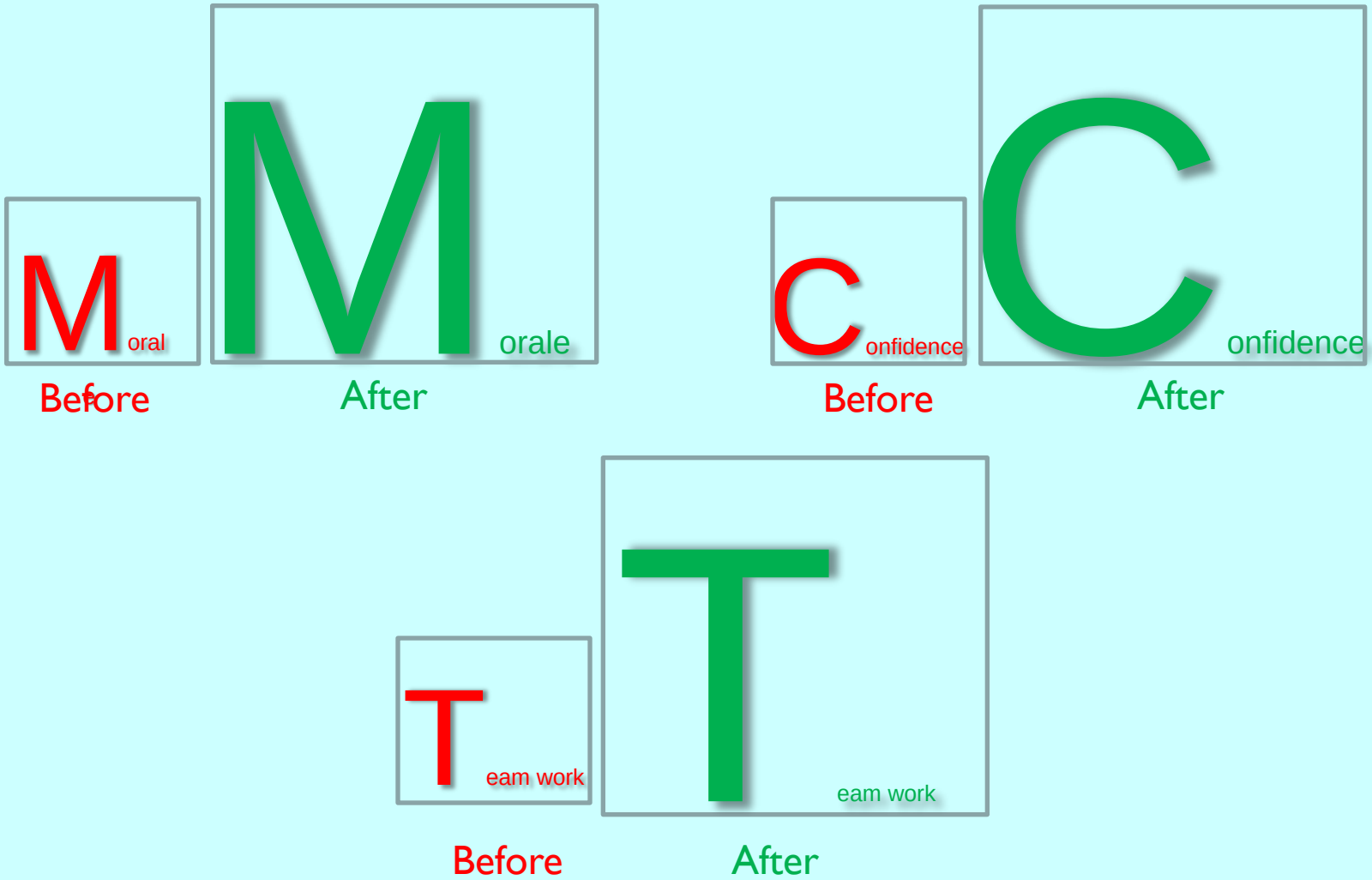


Spoiler mounting hole on rear header outer panel

7.3 Tangible benefits:

Total Annualised Rework Cost Saved = 1,02,000 Rs.

7.3 Intangible benefits:



8. Standardization

8.1 Updation of document

1.Drawing of spoiler mounting hole on rear header outer panel has updated- Sunil Deshmukh – 15.04.2012
2.DCP document has updated for modified hole – Devesh kumar – 15.04.2012
3

8.2 Training

1)Training imparted to side RH operators regarding OK spot on ext. bkt- Mr. Ghosal-25.01.2012-Closed

8.3 Additional Audit or check

1)Checkpoint is added in IQ check sheet to ensure spoiler hole at correct location using checking pin-Mangesh-22.01.2012-Closed

9. Summarize & Plan Future Actions

9.1) Conclusion :

- Spoiler mounting hole provided on rear header has modified from 12mm to 16mm (30.04.2012)

9.2) Horizontal Deployment :

-

9.3) Lessons Learnt :

- In spot welding fixture it is necessary to maintain the fixture parts like clamping ,locating pins to ensure correct assembly.
- There have been delay in implementing the modified spoiler mounting hole on rear header due to tooling cost approval.

9. Summarize & Plan Future Actions

9.1) Conclusion :

- Once target has been achieved, draw implementation plan for remaining identified Causes which cannot get implemented currently.
- If concern doesn't get resolved & Product Design change is necessary, Escalate to PVT/PDT/PET.

9.2) Horizontal Deployment :

- Check the Possibility of Extending the actions to similar product/Process/Model.

9.3) Lessons Learnt :

- Summarize the Learning from this project.
- Mentions the reasons of Delay if any.
- Critique : What has gone Right & What has gone Wrong.

THANK YOU !!!